

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Practical – IV

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AIM: Bayes' Rule Application.

1. Scenario:

A medical test is used to detect a particular disease.
The probability that a randomly selected person has the disease is 1%.
If a person has the disease, the test returns positive with probability 99%.
If a person does not have the disease, the test still returns positive with probability 5% (false positive rate).
A person takes the test and receives a positive result.

2. Problem Statement:

Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

CODE:

```
# Probability of having disease
P_D = 0.01

# Probability of positive test if disease exists (Sensitivity)
P_Pos_D = 0.99

# Probability of positive test if disease does NOT exist (False Positive Rate)
P_Pos_NotD = 0.05

# Probability of not having disease
P_NotD = 1 - P_D

# Bayes' Rule
P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))

# Print result
print("Probability that person actually has the disease:",
      round(P_D_Pos * 100, 2), "%")
```

OUTPUT:

```
...
Probability that person actually has the disease: 16.67 %
```